

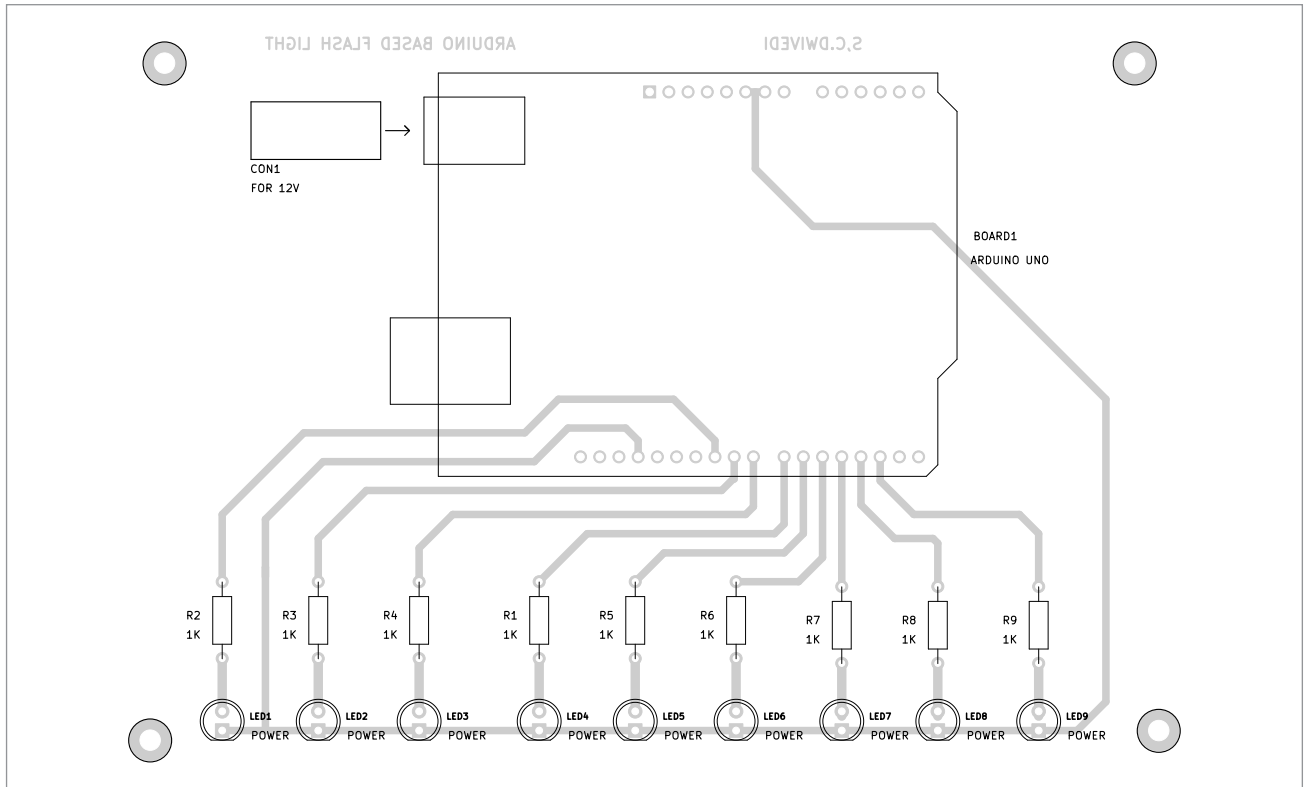


Fig. 5: Components layout for the PCB

of 50ms. So, the pulse switches on only one LED at a time and keeps shifting to the next pin and the LED connected to it.

Before uploading the Sketch program to the Arduino Uno board,

PARTS LIST

Semiconductors:

- Board1 - Arduino board
- LED1-LED9 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R9 - 150-ohm

Miscellaneous:

- CON1 - DC socket
- Breadboard
- Jumper wires

Optional - Parts List for Arduino based 18-LED flashing lights string

Semiconductors:

- Board1 - Arduino board
- LED1-LED18 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R9 - 150-ohm
- R8-R10 - 1-kilo-ohm

Miscellaneous:

- Breadboard
- Jumper wires

do not forget to select the proper port. After selecting proper port, upload the Sketch file flasher.ino to the board.

Construction and testing

In case you wish to make a PCB for the circuit, its same-size pattern is shown in Fig. 4, while the components layout is shown in Fig. 5. After wiring the circuit, enclose it in a suitable box, including the

Arduino Uno board. Solder all the LEDs properly on their footprints (positions) shown in the PCB.

Do not forget to upload the source code (flasher.ino) before assembling the circuit. For uploading the source code use a USB cable, which can also be used to operate this circuit.

If you do not wish to use USB cable to operate the circuit, connect 12V DC from an adaptor across CON1 to operate the circuit.

PCB layout for the circuit in Fig. 3 is not provided here. You can assemble it on a breadboard or a general-purpose. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-arduino-based-flash-light>

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY